

Introduction to Social Data Analytics: Week 4 Part 1

Motivation for Muralidharan, Singh and Ganimian (2019)

- ▶ Many developing countries have rapidly expanded access to education
- ▶ However, not all places see measurable gains in learning
- ▶ For example, in India, over 50 percent of students in grade 5 cannot read at a grade 2 level, despite enrollment rates around 95 percent (Pratham 2017)

Education Mismatch

- ▶ If curriculum is beyond a student's level of understanding, they may learn very little, even if attendance is high
- ▶ Question: can technology that aims to “Teach at the Right Level” be used to improve learning outcomes
- ▶ This is an important question: need to find interventions that work! It is not necessarily true that more investment in education → better outcomes if we are investing in the wrong things

Experiment

- ▶ Mindspark is a computer software that provides learning materials that are appropriate for a student's understanding level
 - ▶ It does this by using information on what questions a student gets right vs. wrong. The level of material only increases in difficulty once the student has learned the previous concepts
- ▶ Mindspark centers provide 6-day-a-week instruction for 90 minutes (45 minutes with Mindspark, 45 minutes from teaching assistants)

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- ▶ Mindspark centers provide 6-day-a-week instruction for 90 minutes (45 minutes with Mindspark, 45 minutes from teaching assistants)
- ▶ Question: Is this program successful in improving learning outcomes?

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- ▶ Participants were recruited from low-income neighborhoods in Delhi
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- ▶ At demonstration sessions for Mindspark, parents were introduced to the program
- ▶ All participants interested in participating needed to fill out a baseline assessment
 - ▶ Then, parents were told that about $\frac{1}{2}$ of participants would receive a voucher that waived tuition for the Mindspark program (participants were chosen by lottery)
 - ▶ Students not chosen by lottery were told they would obtain free access to the centers after February 2016 (about 6 months later)
- ▶ The result is 619 participants, 305 in the control (i.e. no access to Mindspark) and 314 in the treatment (access to Mindspark)

Learning Outcomes

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 - ▶ Baseline: before any student has been offered free Mindspark
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Learning Outcomes

- ▶ We have information about Math and Hindi tests at both the baseline and endline
 - ▶ Baseline: before any student has been offered free Mindspark
 - ▶ Endline: the treated students have been offered free Mindspark
- ▶ Our goal: understand how endline test scores depend on whether you are given access to Mindspark
- ▶ Before we start analyzing the data, however, we are going to learn a new statistical technique: **regression**

Linear Equations

- ▶ A linear equation is any equation that can be written in the following form

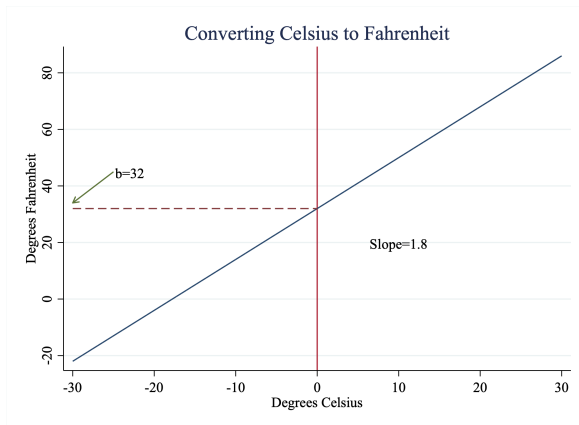
$$y = mx + b \tag{1}$$

- ▶ where m is the slope and b is the intercept

Converting Celsius to Fahrenheit

- There is a linear relationship between degrees in Celsius (C) and degrees in Fahrenheit (F)

$$F = 1.8 \cdot C + 32 \quad (2)$$



Estimating Linear Relationships

- ▶ Often in the social sciences, we are interested in the relationship between two variables
 - ▶ How much does education increase income?
 - ▶ Does precipitation impact voting participation?
- ▶ One way to understand how two variables are related is through estimating linear relationships
- ▶ **Linear Regression** is a way to estimate linear relationships between one or more variables

The Linear Model

- ▶ We want to model some outcome of individuals Y_i as a function of some explanatory variable of the individual X_i ($Y_i = f(X_i)$)
 - ▶ Y_i is often referred to as the dependent variable (or outcome variable)
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 - ▶ Y_i is often referred to as the dependent variable (or outcome variable)
 - ▶ X_i is often referred to as the independent variable (or explanatory variable)
- ▶ If we assume the function $f(X_i)$ is linear, then we can write this model as

$$Y_i = \beta_0 + \beta_1 \cdot X_i + \varepsilon_i$$

A Hypothetical Example

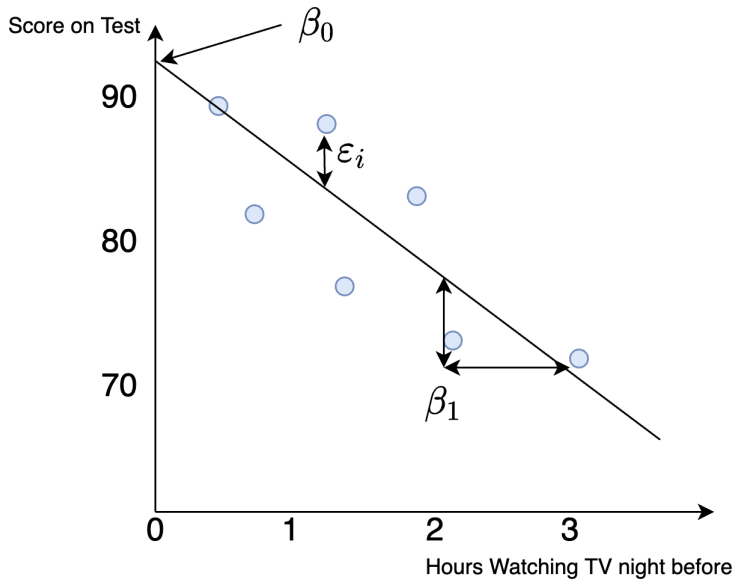
- ▶ Imagine we hypothesize that students who watch too much TV the night before a test perform worse
- ▶ Then we want to understand if test scores (our outcome variable Y_i) is related to hours of TV watched (our explanatory variable X_i)

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- ▶ Imagine we hypothesize that students who watch too much TV the night before a test perform worse
- ▶ Then we want to understand if test scores (our outcome variable Y_i) is related to hours of TV watched (our explanatory variable X_i)
- ▶ We could posit a linear relationship between the two

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad (3)$$

Hypothetical Example



Parameters of the Model

$$Y_i = \beta_0 + \beta_1 \cdot X_i + \varepsilon_i$$

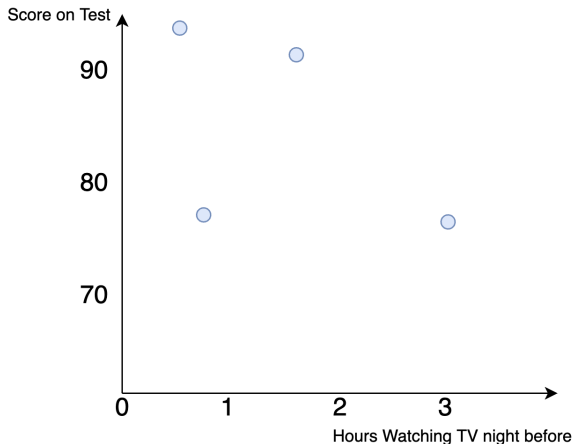
- ▶ We refer to β_0 and β_1 as the parameters of the model
- ▶ We use “hats” to denote the estimates of these parameters
- ▶ $\hat{\beta}_0$ is the estimate of the intercept
- ▶ $\hat{\beta}_1$ is the estimate of the slope coefficient
- ▶ We choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to construct a regression line that best fits our data

Best Fit

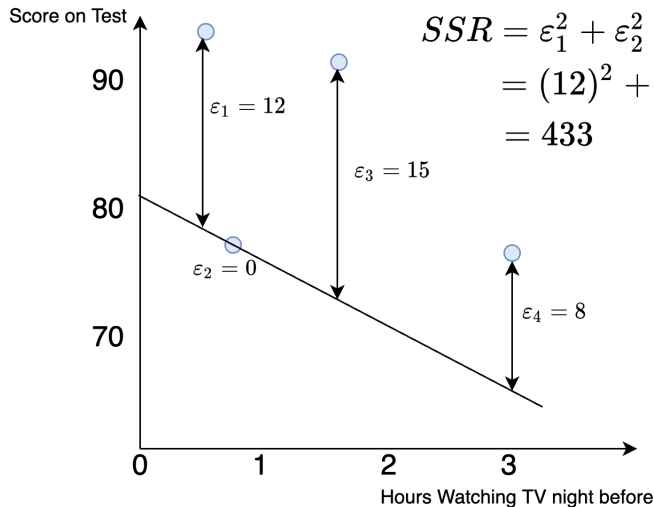
- ▶ What does best fit actually mean?
- ▶ We choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the sum of squared errors ($\sum \varepsilon_i^2$)
- ▶ Choosing the parameters to minimize the sum of squared errors is referred to as **Ordinary Least Squares** and commonly simply referred to as linear regression

Finding the Best Fit

- ▶ How do we go about finding the best fit?
- ▶ Let's consider estimating the relationship between the two variables below

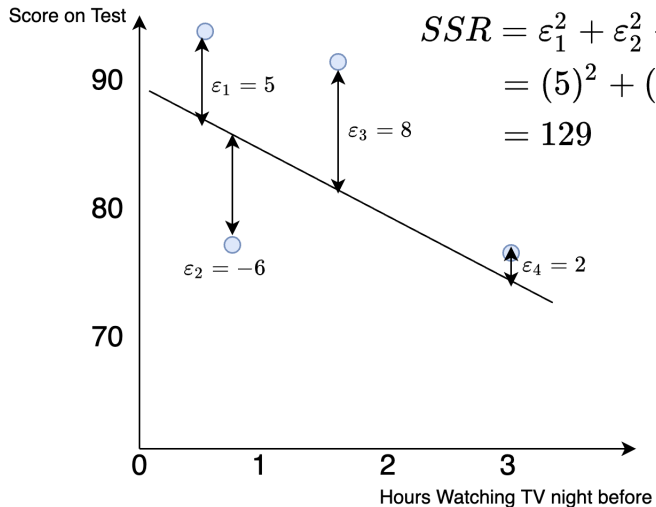


Attempt 1 – Not so Good



$$\begin{aligned} SSR &= \epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 \\ &= (12)^2 + (0)^2 + (15)^2 + (8)^2 \\ &= 433 \end{aligned}$$

Attempt 2 – Much Better



$$\begin{aligned} SSR &= \epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 \\ &= (5)^2 + (-6)^2 + (8)^2 + (2)^2 \\ &= 129 \end{aligned}$$

Estimating a Linear Regression

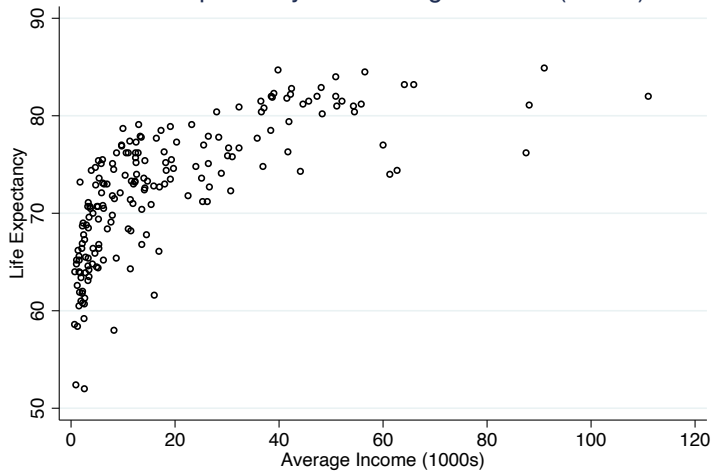
- ▶ Lucky for us, we don't need to actually try out different estimates in practice
- ▶ There is a mathematical solution to this problem (which you will learn about if you take further statistics/econometrics courses in the future)
- ▶ Computers can calculate this solution very fast!

Fitted Values

- ▶ Assume we are interested in the relationship between life expectancy in a country and the average income in a country
- ▶ We propose modeling life expectancy as a linear function of average income

$$\text{Life Expectancy}_i = \beta_0 + \beta_1 \cdot \text{Average Income}_i + \varepsilon_i \quad (4)$$

Life Expectancy and Average Income (1000s)



Estimates

- ▶ We find $\hat{\beta}_0 = 67.97$ and $\hat{\beta}_1 = 0.24$
- ▶ What can we do with this information?
- ▶ One thing we can do is form predictions for life expectancy given a value of average income

Forming Predictions

- ▶ There is sometimes less emphasis on forming predictions in social sciences (we will discuss why soon)
- ▶ However, in many settings, one might be interested in forming predictions
 - ▶ Industry: a company may try to predict who will buy their good based on characteristics of individuals
 - ▶ Policy: a policymaker may want to predict some outcomes of individuals, such as predicting recidivism for incarcerated individuals
- ▶ Linear regression is one (powerful) way to form predictions. It is the first tool you will learn in most machine learning classes

Forming Predictions

- ▶ Imagine I tell you a country has an average income of \$50k
- ▶ How do we use the regression results to form our best guess for the life expectancy in this country?

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$$\text{Predicted } Y_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot X_i \quad (5)$$

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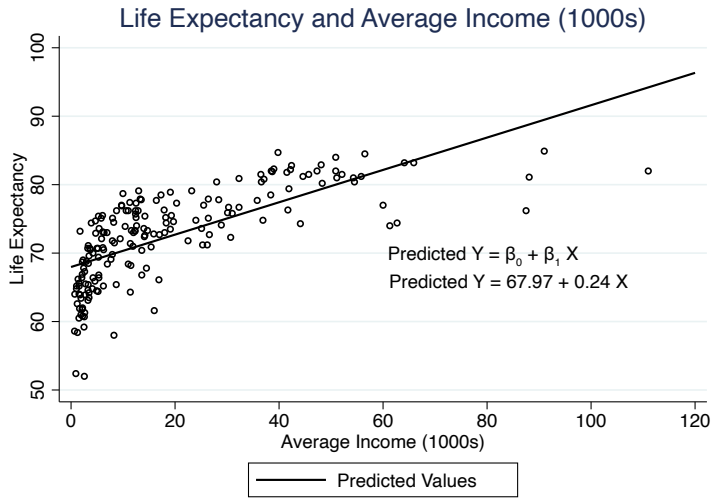
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$$\text{Predicted } Y_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot X_i \quad (5)$$

- ▶ In this particular application

$$\text{Predicted Life Expectancy}_i = 67.97 + 0.24 \cdot 50 = 79.97 \quad (6)$$

Visualizing the Regression



The Slope Coefficient

- ▶ To remind you, the linear regression model is given by:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- ▶ As social scientists, we are often interested how changes in our explanatory variable (X variable) are associated with changes in our outcome variable (Y variable)
- ▶ The slope coefficient (β_1) is informative about this

Life Expectancy and Average Income

- Recall that we modeled Life Expectancy as a linear function of Average Income (1000s) and found

$$\text{Predicted Life Expectancy}_i = 67.97 + 0.24 \cdot \text{Average Income}_i$$

- The slope coefficient captures how changes in Average Income are associated with changes in Predicted Life Expectancy

The Slope Coefficient – Example

- ▶ Imagine I tell you I have two countries, Country A and Country B
- ▶ Country A has an Average Income of \$50k
- ▶ Country B has an Average Income of \$51k
- ▶ What is the expected difference in Life Expectancy between these two countries?

The Slope Coefficient – Example

- ▶ We can form the Predicted Life Expectancy in both of these countries

$$\text{Predicted Life Expectancy A} = 67.97 + 0.24 \cdot 50$$

$$\text{Predicted Life Expectancy B} = 67.97 + 0.24 \cdot 51$$

The Slope Coefficient – Example

- ▶ We can form the Predicted Life Expectancy in both of these countries

$$\text{Predicted Life Expectancy A} = 67.97 + 0.24 \cdot 50$$

$$\text{Predicted Life Expectancy B} = 67.97 + 0.24 \cdot 51$$

- ▶ Taking the difference between these two yields

$$\text{Difference in Predicted Life Expectancy} = 0.24 = \text{slope coefficient}$$

The Slope Coefficient

- ▶ The slope coefficient is generally reported in order to understand the magnitude of a relationship
- ▶ General interpretation:
 - ▶ A 1-unit change in X is associated with a **slope-coefficient** unit change in the Y variable

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 - ▶ A 1-unit change in X is associated with a **slope-coefficient** unit change in the Y variable
- ▶ In our example
 - ▶ A 1-unit change in Average Income (1000s) is associated with 0.24 year increase in Life Expectancy

The reg Command

- ▶ Let's mathematically set up a general linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- ▶ The basic syntax for the reg command is

```
reg yvar xvar
```

- ▶ This will produce a table with many statistics. For now we will focus on the intercept $\hat{\beta}_0$ and slope $\hat{\beta}_1$

Using the reg Command

- ▶ Let's load the data on life expectancy and try our first regression

```
/*load data*/  
cd "/Users/davidarnold/Dropbox/Teaching/EP5/online/04_week/data"  
use gapminder.dta, replace
```

/Users/davidarnold/Dropbox/Teaching/EP5/online/04_week/data

- ▶ Reminder, we are modeling life expectancy as a function of income per capita (in the thousands)
- ▶ This means our dependent variable (or Y-variable) will be life expectancy and our independent variable (or X-variable) will be income per capita (in the thousands)

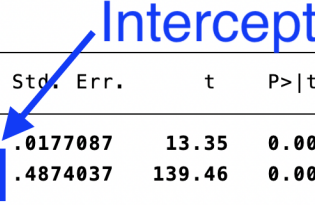
The reg Command

```
/*regression of life expectancy on average income per capita*/  
reg life_expectancy average_income
```

The Intercept

Source	SS	df	MS	Number of obs	=	186
				F(1, 184)	=	178.14
Model	4150.14525	1	4150.14525	Prob > F	=	0.0000
Residual	4286.67682	184	23.2971567	R-squared	=	0.4919
				Adj R-squared	=	0.4891
Total	8436.82208	185	45.6044436	Root MSE	=	4.8267

life_expecta~y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
average_income	.2363563	.0177087	13.35	0.000	.201418	.2712945
_cons	67.97336	.4874037	139.46	0.000	67.01174	68.93498

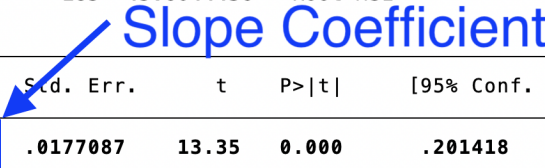


- The intercept is also referred to as the constant, which is why it appears next to something labeled `_cons`

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- The slope coefficient for a given variable always appears next to the variable name

Visualizing the Regression

- ▶ You should be able to draw the regression line from this output manually (i.e. if given pen and paper, figure out what the regression line looks like)
- ▶ Intercept – At ($X=0$), or zero average income, the regression line is equal to 67.97
- ▶ Slope – Every 1 unit increase in average income (which is denoted in the thousands), there is an associated 0.24 increase in average life expectancy

Plotting the Regression Line in Stata

- ▶ Stata can plot the regression line for you through the `twoway lfit` command
 - ▶ Other terms for linear regression line (line-of-best fit, predicted values)
- ▶ The basic syntax is

```
twoway lfit yvar xvar
```

- ▶ This will plot the linear regression line from the output of

```
reg yvar xvar
```

- If there are not too many observations (say less than 1000), can also plot scatter along with regression line

```
twoway scatter life_expectancy average_income, ///  
    msymbol(circle_hollow) msize(small) ///  
    || lfit life_expectancy average_income, lw(0.4) lc(red) ///  
    title("Regression of Life Expectancy on Average Income") ///  
    xtitle("Average Income (1000s)") ///  
    ytitle("Life Expectancy") ///  
    graphregion(color(white) fcolor(white))
```

